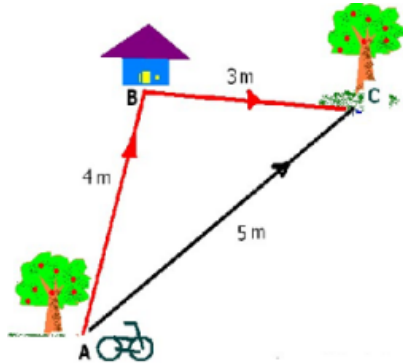


MOTION-05

Class 09 - Science

1. The path covered from A to C is called:

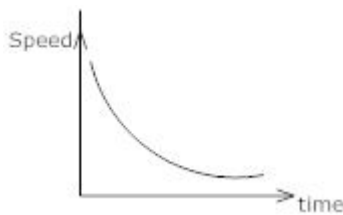
[1]



- a) Velocity b) Speed
c) Distance d) Displacement

2. The below graph represents

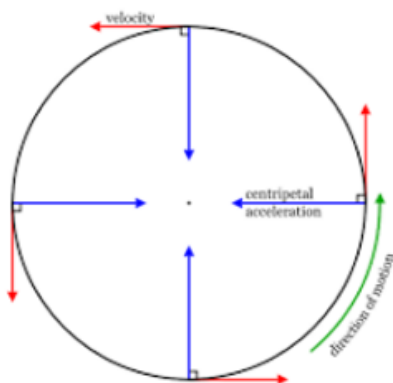
[1]



- a) constant retardation b) non – uniform retardation
c) constant acceleration d) constant velocity

3. The given image shows the motion of the moon around the earth. Which of the following statement/s is/are correct regarding the given image?

[1]

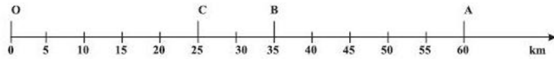


- a) The velocity of an object changes b) All of these
- c) The speed of the object remains constant d) A body moves in a circle

4. If the object moves through C and B and reaches A. Then it moves back along the same path and reaches C

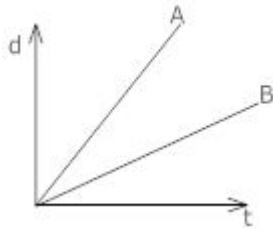
[1]

through B. What is the total path covered?



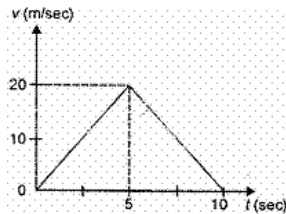
- a) 105 Km
- b) 95 Km
- c) 65 Km
- d) 35 Km

5. Which of the following is correct about the following distance-time graph? [1]



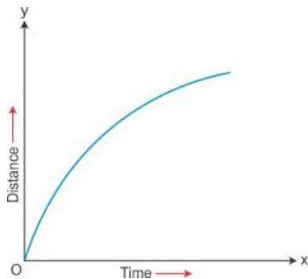
- a) speed of A is equal to speed of B
- b) None of these.
- c) speed of A is greater than speed of B
- d) speed of A is less than speed of B

6. The displacement of the body after 7.5 seconds from the beginning of the motion is [1]



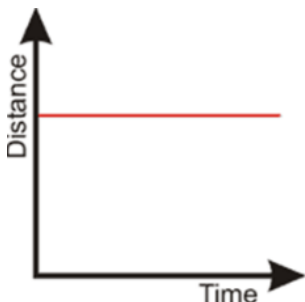
- a) 12.5 m
- b) 75 m
- c) 100 m
- d) 87.5 m

7. The given image shows a graph of distance travelled against time is a curved line. This graph is meant for: [1]



- a) Uniform acceleration
- b) Uniform speed
- c) Non-uniform speed
- d) Non-uniform acceleration

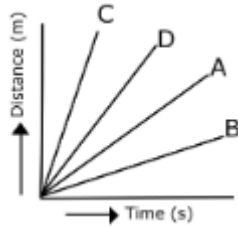
8. According to the given image of a Distance-Time graph, the speed of a moving body: [1]



- a) Decreases
- b) First increases and then decreases
- c) Increases
- d) Becomes zero

9. Four cars A, B, C and D are moving on a levelled road. Their distance-time graph is given as under. The correct [1]

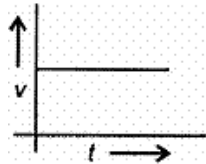
statement is:



- a) Car B is slowest
- b) Car C is slowest
- c) Car D is faster than Car C
- d) Car A is slowest

10. From the given v-t graph, it can be inferred that the object is:

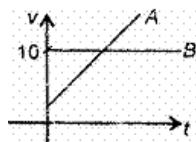
[1]



- a) in non-uniform motion
- b) at rest
- c) Non of these
- d) in uniform motion

11. The v-t graph shown here depicts the motion of A and B such that

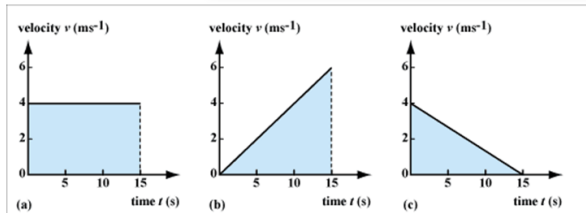
[1]



- a) they collide when their velocity is 10 ms^{-1}
- b) both A and B have zero acceleration
- c) both A and B have non-zero acceleration
- d) velocity of A exceeds beyond 10 ms^{-1}

12. Which of the Velocity – Time graph shows uniform retardation?

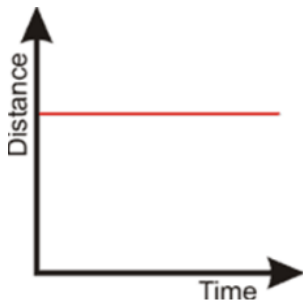
[1]



- a) b
- b) a and b
- c) c
- d) a

13. The given image of Distance – Time graph represents that

[1]

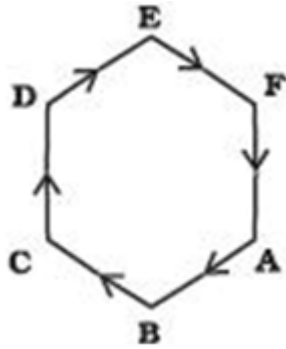


- a) a body is travelling with non-uniform speed
- b) None of these
- c) a body stops moving
- d) a body is travelling with uniform speed

14. If the athlete is running along the shown path ABCDEF, the athlete will have to change his direction six times

[1]

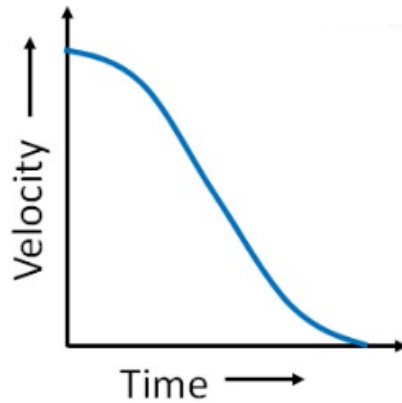
while he completes one round. He is running on which path?



- a) Hexagonal path
- b) Trigonal path
- c) Heptagonal path
- d) Pentagonal path

15. The graph shown in the image represents:

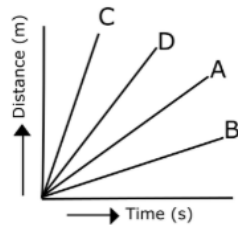
[1]



- a) Non-uniform retardation
- b) Uniform retardation
- c) Non-uniform acceleration
- d) Uniform acceleration

16. Four cars A, B, C and D are moving on the levelled road. Their distance-time graph is given as under. The correct statement is:

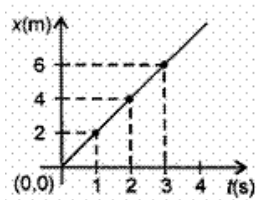
[1]



- a) Car D is faster than Car C
- b) Car A is faster than car D
- c) Car C is slowest
- d) Car B is slowest

17. The slope of the x-t graph is a measure of

[1]



- a) velocity = 2ms^{-1}
- b) acceleration = 2ms^{-2}
- c) acceleration = $\frac{1}{2}\text{ms}^{-2}$
- d) velocity = $\frac{1}{2}\text{ms}^{-2}$

18. Match the following with the correct response.

[1]

(a) Graph showing uniform velocity	(i) Straight-line sloping upwards
------------------------------------	-----------------------------------

(b) Graph showing uniformly accelerated velocity	(ii) Curved line
(c) Graph showing uniformly retarded	(iii) Straight-line sloping downwards
(d) Graph showing non-uniform motion	(iv) Straight-line parallel to x-axis

a) (a) - (i), (b) - (iii), (c) - (ii), (d) - (iv)

b) (a) - (iii), (b) - (ii), (c) - (iv), (d) - (i)

c) (a) - (iv), (b) - (i), (c) - (iii), (d) - (ii)

d) (a) - (ii), (b) - (iv), (c) - (i), (d) - (iii)

19. Match the following with the correct response:

[1]

(a) Graph showing uniform velocity	(i) Straight-line sloping upwards
(b) Graph showing uniformly accelerated velocity	(ii) Curved line
(c) Graph showing uniformly retarded	(iii) Straight-line sloping downwards
(d) Graph showing non-uniform motion	(iv) Straight-line parallel to the x-axis

a) (a) - (ii), (b) - (iv), (c) - (i), (d) - (iii)

b) (a) - (i), (b) - (iii), (c) - (ii), (d) - (iv)

c) (a) - (iv), (b) - (i), (c) - (iii), (d) - (ii)

d) (a) - (iii), (b) - (ii), (c) - (iv), (d) - (i)

20. Match the following with the correct response.

[1]

(a) Non- uniform motion	(i) The slope of the graph
(b) Speed from the distance-time graph	(ii) $\frac{\text{Initial velocity} + \text{Final velocity}}{2}$
(c) Distance covered from the speed-time graph	(iii) The area under the speed-time graph
(d) Average velocity	(iv) If a body covers unequal distances in equal intervals of time.

a) (a) - (ii), (b) - (iv), (c) - (i), (d) - (iii)

b) (a) - (iv), (b) - (i), (c) - (iii), (d) - (ii)

c) (a) - (i), (b) - (iii), (c) - (ii), (d) - (iv)

d) (a) - (iii), (b) - (ii), (c) - (iv), (d) - (i)

21. Match the following with the correct response.

[1]

(a) I st equation of motion	(i) $v^2 = u^2 + 2aS$
(b) II nd equation of motion	(ii) $v = u + at$
(c) III rd equation of motion	(iii) $-\left(\frac{v-u}{t}\right)$
(d) Retardation	(iv) $S = ut + \frac{1}{2}at^2$

a) (a) - (ii), (b) - (iv), (c) - (i), (d) - (iii)

b) (a) - (i), (b) - (iii), (c) - (ii), (d) - (iv)

c) (a) - (iv), (b) - (i), (c) - (iii), (d) - (ii)

d) (a) - (iii), (b) - (ii), (c) - (iv), (d) - (i)

22. Match the following with correct response.

[1]

Column A	Column B
(1) Velocity	(A) m/s^2
(2) Displacement	(B) m/s

(3) Acceleration	(C) $-m/s^2$
(4) Retardation	(D) m

a) 1-D, 2-A, 3-C, 4-B

b) 1-B, 2-D, 3-A, 4-C

c) 1-C, 2-B, 3-D, 4-A

d) 1-A, 2-C, 3-B, 4-D

23. Match the following with the correct response.

[1]

(a) v	(i) $\frac{v-u}{t}$
(b) a	(ii) $v^2 - u^2$
(c) V_{av}	(iii) $\frac{v+u}{2}$
(d) $2aS$	(iv) $\frac{s}{t}$

a) (a) - (iii), (b) - (ii), (c) - (iv), (d) - (i)

b) (a) - (ii), (b) - (iv), (c) - (i), (d) - (iii)

c) (a) - (iv), (b) - (i), (c) - (iii), (d) - (ii)

d) (a) - (i), (b) - (iii), (c) - (ii), (d) - (iv)

24. Match the following with correct response.

[1]

Column A	Column B
(1) Non- uniform motion	(A) $\frac{\text{Final velocity} + \text{Initial velocity}}{2}$
(2) Speed from distance-time graph	(B) Slope of the graph
(3) Distance covered from speed time graph	(C) If a body covers unequal distances in equal intervals of time
(4) Average velocity	(D) Area under speed time graph

a) 1-A, 2-C, 3-B, 4-D

b) 1-D, 2-A, 3-C, 4-B

c) 1-C, 2-B, 3-D, 4-A

d) 1-B, 2-D, 3-A, 4-C

25. Match the following with correct response.

[1]

(1) Speed	(A) $\frac{\text{Displacement}}{\text{Time taken}}$
(2) Velocity	(B) $\frac{\text{total distance covered}}{\text{total time taken}}$
(3) Acceleration	(C) $\frac{\text{Change in velocity}}{\text{Time taken}}$
(4) Average speed	(D) $\frac{\text{Distance travelled}}{\text{Time take}}$

a) 1-B, 2-D, 3-A, 4-C

b) 1-D, 2-A, 3-C, 4-B

c) 1-C, 2-B, 3-D, 4-A

d) 1-A, 2-C, 3-B, 4-D

26. Match the following with correct response.

[1]

(1) Speed	(A) $\frac{\text{Change of velocity}}{\text{Time taken}}$
(2) Velocity	(B) $\frac{\text{Distance covered}}{\text{Time taken}}$
(3) Acceleration	(C) $\frac{\text{Total distance covered}}{\text{Total time taken}}$
(4) Average speed	(D) $\frac{\text{Displacement}}{\text{Time taken}}$

a) 1-A, 2-C, 3-B, 4-D

b) 1-C, 2-B, 3-D, 4-A

c) 1-B, 2-D, 3-A, 4-C

d) 1-D, 2-A, 3-C, 4-B

27. Match the following with the correct response.

[1]

(a) Rate of change of velocity	(i) Uniform circular motion
(b) Rate of change of displacement	(ii) Velocity
(c) Rate of change of distance	(iii) Acceleration
(d) Rate of change of speed in a circular path	(iv) Speed

a) (a) - (iv), (b) - (i), (c) - (iii), (d) - (ii)

b) (a) - (i), (b) - (iii), (c) - (ii), (d) - (iv)

c) (a) - (iii), (b) - (ii), (c) - (iv), (d) - (i)

d) (a) - (ii), (b) - (iv), (c) - (i), (d) - (iii)

28. Match the following with the correct response:

[1]

(a) Velocity	(i) m/s^2
(b) Displacement	(ii) m/s
(c) Acceleration	(iii) s
(d) Time	(iv) m

a) (a) - (i), (b) - (iii), (c) - (ii), (d) - (iv)

b) (a) - (ii), (b) - (iv), (c) - (i), (d) - (iii)

c) (a) - (iii), (b) - (ii), (c) - (iv), (d) - (i)

d) (a) - (iv), (b) - (i), (c) - (iii), (d) - (ii)

29. Match the following with the correct response:

[1]

(a) I equation of motion	(i) $v = u + at$
(b) II equation of motion	(ii) $v^2 = u^2 + 2as$
(c) III equation of motion	(iii) $S = ut + \frac{1}{2}at^2$
(d) Retardation	(iv) $-\left(\frac{v-u}{t}\right)$

a) (a) - (i), (b) - (iii), (c) - (ii), (d) - (iv)

b) (a) - (iii), (b) - (ii), (c) - (iv), (d) - (i)

c) (a) - (iv), (b) - (i), (c) - (iii), (d) - (ii)

d) (a) - (ii), (b) - (iv), (c) - (i), (d) - (iii)

30. Match the following with the correct response.

[1]

(a) Uniform acceleration	(i) Vector quantity
(b) Measurement of distance	(ii) Odometer
(c) Energy	(iii) Velocity increases in equal intervals of time
(d) Acceleration	(iv) Scalar quantity

a) (a) - (ii), (b) - (iv), (c) - (i), (d) - (iii)

b) (a) - (iv), (b) - (i), (c) - (iii), (d) - (ii)

c) (a) - (i), (b) - (iii), (c) - (ii), (d) - (iv)

d) (a) - (iii), (b) - (ii), (c) - (iv), (d) - (i)

31. Match the following with the correct response:

[1]

(a) Uniform acceleration	(i) Velocity increases in equal intervals of time
(b) Measurement of distance	(ii) Scalar quantity
(c) Energy	(iii) Odometer

(d) Acceleration	(iv) Vector quantity
------------------	----------------------

a) (a) - (i), (b) - (iii), (c) - (ii), (d) - (iv)

b) (a) - (iii), (b) - (ii), (c) - (iv), (d) - (i)

c) (a) - (ii), (b) - (iv), (c) - (i), (d) - (iii)

d) (a) - (iv), (b) - (i), (c) - (iii), (d) - (ii)

32. Match the following with the correct response:

[1]

(a) Scalar	(i) Magnitude and direction
(b) Vector	(ii) Motion of satellite
(c) Rectilinear motion	(iii) straight-line motion
(d) Circular motion	(iv) Magnitude

a) (a) - (iii), (b) - (ii), (c) - (iv), (d) - (i)

b) (a) - (ii), (b) - (iv), (c) - (i), (d) - (iii)

c) (a) - (iv), (b) - (i), (c) - (iii), (d) - (ii)

d) (a) - (i), (b) - (iii), (c) - (ii), (d) - (iv)

33. Match the following with correct response.

[1]

Column A	Column B
(1) v	(A) $\frac{v-u}{t}$
(2) a	(B) $v^2 - u^2$
(3) V_{av}	(C) $\frac{v+u}{2}$
(4) $2aS$	(D) $\frac{s}{t}$

a) 1-A, 2-C, 3-B, 4-D

b) 1-C, 2-B, 3-D, 4-A

c) 1-B, 2-D, 3-A, 4-C

d) 1-D, 2-A, 3-C, 4-B

34. Match the following with the correct response:

[1]

(a) Rate of change of velocity	(i) Uniform circular motion
(b) Rate of change of displacement	(ii) Velocity
(c) Rate of change of distance	(iii) Acceleration
(d) Rate of change of speed in a circular path	(iv) Speed

a) (a) - (i), (b) - (iii), (c) - (ii), (d) - (iv)

b) (a) - (iii), (b) - (ii), (c) - (iv), (d) - (i)

c) (a) - (iv), (b) - (i), (c) - (iii), (d) - (ii)

d) (a) - (ii), (b) - (iv), (c) - (i), (d) - (iii)

35. Match the following with the correct response.

[1]

(a) Scalar	(i) Magnitude and direction
(b) Vector	(ii) Motion of satellite
(c) Rectilinear motion	(iii) Non-uniform
(d) Circular motion	(iv) Magnitude

a) (a) - (i), (b) - (iii), (c) - (ii), (d) - (iv)

b) (a) - (iv), (b) - (i), (c) - (iii), (d) - (ii)

c) (a) - (ii), (b) - (iv), (c) - (i), (d) - (iii)

d) (a) - (iii), (b) - (ii), (c) - (iv), (d) - (i)

36. Distance can never be _____ for a moving body.

[1]

37. Distance is a _____ quantity. [1]

a) Positive
b) Negative or zero
c) Zero
d) Negative

38. The equation of motion is applicable for _____ acceleration motion. [1]

a) Vector
b) Unit vector
c) Dimensionless
d) Scalar

39. The maximum displacement between two points in a circular path can be _____. (Radius-r) [1]

a) $2r$
b) r
c) πr
d) $2\pi r$

40. **Statement A:** When a body moves with uniform speed, then the average speed is the same as instantaneous speed.
Statement B: Average speed is less than the average velocity.
Which of the above statement is true?

a) Statement B
b) Neither A nor B
c) Statement A
d) Both A and B

41. Which of the statement is true? [1]

Statement A: A curved line in a speed-time graph means non-uniform motion.
Statement B: A curved line in a velocity-time graph means uniform acceleration.

a) Statement A is false, B is true
b) Both the statement A and B are true
c) Neither statement A nor Statement B is true.
d) Statement A is true, B is false

42. **Statement A:** Uniform circular motion is a case of accelerated motion. [1]
Statement B: In the third equation of motion we do not have the time.
Which of the following statements are correct:

a) Neither statement A nor statement B is true.
b) Statement B is true, A is false
c) Both statements A and B are true
d) Statement A is true, B is false